

First progress-tracking pilot

psi_0.5_r_abc_v2, checkpoint 820000. ABC mug flipping: 4 development and 8 test episodes, 12 timestamps each.

Method	Completion	MAE ↓	Raw progress	Spearman ↑
Constant	0.328	—	—	—
DT	0.327	0.181	—	—
S1	0.332	0.084	—	—
S3	0.331	0.070	—	—
S5	0.323	0.154	—	—
Elapsed time	0.196	0.935	—	—
DT + S1	0.329	—	—	—
DT + S3	0.327	—	—	—
DT + S5	0.321	—	—	—

Pilot finding: model-based completion errors remain close to the constant baseline; elapsed time is substantially stronger on these successful demonstrations. DT + S5 is numerically best among model-based methods, but its paired difference from DT has a 95% interval spanning zero. There is no reliable combination gain in this pilot. Independent-goal appearance/layout mismatch is a possible confound, not an established explanation.

S1/S3/S5 mean negative goal surprisal with 1/3/5 causal top-camera observations. DT is negative expected goal distance over 4096 bins. All learned calibrations use only the four development episodes.

- One task; eight test episodes; all successful demonstrations.
- Independent reference has different mug colors and randomized layout; scene matching is incomplete.
- 128 fixed shuffled patches/image; controlled history comparison differs from the training uniform total-budget allocation.
- Ridge fits can reverse raw-score signs; raw correlation and coefficients are reported.
- Training membership not fully audited; no zero-shot generalization claim.
- DIN0v2, Robometer, bottles and plates have not been run.

All episodes ended successfully: terminal success/failure AUROC is unavailable. Normalized time is not used as a deployable baseline; the elapsed-time baseline receives only seconds since episode start.

Protocol

```
{
  "seed": 20260919,
  "task": "sim_turn the mug_right_side up",
  "checkpoint": "/ccn2b/dataset/brd/checkpoints_atlask/psi_0.5_r_abc_v2/model_00820000",
  "goal": {
    "split": "val",
    "episode_id": "episode_019e3937-3fda-7f73-a7ea-3333280a53a1",
    "frames": 809,
    "mug_qpos_addresses": [
      0,
      7
    ],
    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "goal",
    "frame_indices": [
      808
    ]
  },
  "episodes": [
    {
      "split": "val",
      "episode_id": "episode_019e270e-c446-7199-8439-6431c74daad3",
      "frames": 709,
    }
  ]
}
```

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    "mug_qpos_addresses": [
      0,
      7
    ],
    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "dev",
    "frame_indices": [
      24,
      84,
      144,
      210,
      270,
      330,
      396,
      456,
      516,
      582,
      642,
      708
    ],
    "reward": [
      0.0,
      0.0,
      0.0,
      0.0,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      1.0,
      1.0,
      1.0
    ]
  },
  {
    "split": "val",
    "episode_id": "episode_019e2061-8855-7cf1-bb20-a64ffac153ec",
    "frames": 1281,
    "mug_qpos_addresses": [
      0,
      7
    ],
    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "dev",
    "frame_indices": [
      24,
      138,
      252,
      366,
      480,
      594,
      708,
      822,
      936,
      1050,
      1164,
      1278
    ],
  },

```

```

"reward": [
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.5,
  0.5,
  0.5,
  1.0,
  1.0
]
},
{
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  "episode_id": "episode_019e258a-7513-71b7-adff-4bfea63390e6",
  "frames": 791,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "dev",
  "frame_indices": [
    24,
    90,
    162,
    228,
    300,
    366,
    438,
    504,
    576,
    642,
    714,
    786
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0
  ]
},
{
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  "episode_id": "episode_019e3d01-3a03-706c-84e2-61080e75aa7b",
  "frames": 767,
  "mug_qpos_addresses": [
    0,
    7
  ],

```

```

    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "dev",
    "frame_indices": [
      24,
      90,
      156,
      222,
      288,
      354,
      426,
      492,
      558,
      624,
      690,
      762
    ],
    "reward": [
      0.0,
      0.0,
      0.0,
      0.0,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      1.0,
      1.0,
      1.0
    ]
  },
  {
    "split": "val",
    "episode_id": "episode_019e3c1d-21c6-7f26-a06b-d82de993e049",
    "frames": 788,
    "mug_qpos_addresses": [
      0,
      7
    ],
    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "test",
    "frame_indices": [
      24,
      90,
      162,
      228,
      300,
      366,
      438,
      504,
      576,
      642,
      714,
      786
    ],
    "reward": [
      0.0,
      0.0,
      0.0,

```

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0.0,
0.5,
0.5,
0.5,
0.5,
0.5,
1.0,
1.0,
1.0
]
},
{
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  "episode_id": "episode_019e3bfb-1ac2-7fd0-916c-12998d58fa5f",
  "frames": 768,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    90,
    156,
    222,
    288,
    354,
    426,
    492,
    558,
    624,
    690,
    762
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0,
    1.0
  ]
},
{
  "split": "val",
  "episode_id": "episode_019e2a8d-69f8-7a1a-a3da-afbfb77b1682",
  "frames": 1073,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,

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"role": "test",
"frame_indices": [
  24,
  114,
  210,
  306,
  402,
  498,
  588,
  684,
  780,
  876,
  972,
  1068
],
"reward": [
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  1.0,
  1.0
]
},
{
  "split": "val",
  "episode_id": "episode_019e2262-668e-7921-b3c7-e9a6cf932323",
  "frames": 651,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    78,
    132,
    192,
    246,
    306,
    360,
    420,
    474,
    534,
    588,
    648
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,

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```

    0.5,
    0.5,
    1.0,
    1.0,
    1.0
  ],
},
{
  "split": "val",
  "episode_id": "episode_019e26fc-ddf4-70ac-9383-698200cac6b3",
  "frames": 663,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    78,
    138,
    192,
    252,
    312,
    366,
    426,
    486,
    540,
    600,
    660
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0,
    1.0
  ]
},
{
  "split": "val",
  "episode_id": "episode_019e27bf-8328-73db-8263-d9e213c96c49",
  "frames": 623,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    78,

```

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132,
186,
240,
294,
348,
402,
456,
510,
564,
618
],
"reward": [
0.0,
0.0,
0.0,
0.0,
0.0,
0.5,
0.5,
0.5,
0.5,
1.0,
1.0,
1.0
]
},
{
"split": "val",
"episode_id": "episode_019e1d6a-4038-7013-bffe-e69df784231a",
"frames": 616,
"mug_qpos_addresses": [
0,
7
],
"nq": 30,
"num_mugs": 2,
"final_reward": 1.0,
"ever_success": true,
"role": "test",
"frame_indices": [
24,
72,
126,
180,
234,
288,
342,
396,
450,
504,
558,
612
],
"reward": [
0.0,
0.0,
0.0,
0.0,
0.5,
0.5,
0.5,
0.5,
0.5,
1.0,
1.0,
1.0,

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    1.0
  ],
},
{
  "split": "val",
  "episode_id": "episode_019e2891-b03f-7e80-a987-2e2e8431a80e",
  "frames": 703,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    84,
    144,
    204,
    270,
    330,
    390,
    450,
    516,
    576,
    636,
    702
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0
  ]
}
],
"histories": [
  1,
  3,
  5
],
"patches_per_frame": 128,
"target_offset_slots": 5,
"fps": 5,
"image_hw": [
  256,
  336
],
"label_rule": "fraction of mugs upright_z >= .95 and height <= first_frame_height + .03 m",
"scope": "single-task pilot; independent successful goal; known training overlap audit only excludes bottle data, no zero-shot claim"
}

Metrics and calibration
{

```

```

"checkpoint_step": 820000,
"task": "mug flipping",
"dev_episodes": 4,
"test_episodes": 8,
"timestamps_per_episode": 12,
"methods": {
  "Constant": {
    "test_episode_macro_mae": 0.328125,
    "mae_95ci": [
      0.31857638888888884,
      0.33854166666666667
    ],
    "test_episode_macro_spearman": null,
    "valid_correlation_episodes": 0,
    "raw_score_spearman": null,
    "features": [],
    "standardization_mean": [],
    "standardization_std": [],
    "coefficients": []
  },
  "DT": {
    "test_episode_macro_mae": 0.3273405337038059,
    "mae_95ci": [
      0.3075229124362797,
      0.34897135853279365
    ],
    "test_episode_macro_spearman": 0.18112746189229073,
    "valid_correlation_episodes": 8,
    "raw_score_spearman": 0.18112746189229073,
    "features": [
      "DT"
    ],
    "standardization_mean": [
      -197.50550079345703
    ],
    "standardization_std": [
      71.60835921260684
    ],
    "coefficients": [
      0.3958333333333333,
      0.11113108350125962
    ]
  },
  "S1": {
    "test_episode_macro_mae": 0.3319016533156805,
    "mae_95ci": [
      0.3170350421945455,
      0.34604337328240264
    ],
    "test_episode_macro_spearman": 0.08405835734680098,
    "valid_correlation_episodes": 8,
    "raw_score_spearman": 0.08405835734680098,
    "features": [
      "S1"
    ],
    "standardization_mean": [
      -9.691885570685068
    ],
    "standardization_std": [
      0.4256916108194747
    ],
    "coefficients": [
      0.39583333333333335,
      0.11137074640919019
    ]
  }
}

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},
"S3": {
  "test_episode_macro_mae": 0.3310623249560374,
  "mae_95ci": [
    0.30745231038595394,
    0.3519668889851141
  ],
  "test_episode_macro_spearman": 0.07048287697140052,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.07048287697140052,
  "features": [
    "S3"
  ],
  "standardization_mean": [
    -9.528732657432556
  ],
  "standardization_std": [
    0.4013215544248379
  ],
  "coefficients": [
    0.3958333333333333,
    0.1214514875874025
  ]
},
"S5": {
  "test_episode_macro_mae": 0.3234628925934101,
  "mae_95ci": [
    0.2971959419081484,
    0.35094781615582554
  ],
  "test_episode_macro_spearman": 0.15386754529173968,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.15386754529173968,
  "features": [
    "S5"
  ],
  "standardization_mean": [
    -9.386288225650787
  ],
  "standardization_std": [
    0.39092901221649673
  ],
  "coefficients": [
    0.3958333333333333,
    0.14487883394707893
  ]
},
"Elapsed time": {
  "test_episode_macro_mae": 0.19623382293657682,
  "mae_95ci": [
    0.17311442861251197,
    0.2185598614881379
  ],
  "test_episode_macro_spearman": 0.9353970709613129,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.9353970709613129,
  "features": [
    "Elapsed time"
  ],
  "standardization_mean": [
    15.0625
  ],
  "standardization_std": [
    10.06935584252207
  ]
},

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"coefficients": [
  0.3958333333333333,
  0.27525928339574546
],
},
"DT + S1": {
  "test_episode_macro_mae": 0.32875687786060204,
  "mae_95ci": [
    0.307056029370113,
    0.35278785518193945
  ],
  "test_episode_macro_spearman": 0.11145697462927062,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": null,
  "features": [
    "DT",
    "S1"
  ],
  "standardization_mean": [
    -197.50550079345703,
    -9.691885570685068
  ],
  "standardization_std": [
    71.60835921260684,
    0.42569161081947465
  ],
  "coefficients": [
    0.3958333333333335,
    0.08141515757414476,
    0.08179157930807485
  ]
},
"DT + S3": {
  "test_episode_macro_mae": 0.3271711423834641,
  "mae_95ci": [
    0.3036968790559315,
    0.35500832006881305
  ],
  "test_episode_macro_spearman": 0.15145360923427686,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": null,
  "features": [
    "DT",
    "S3"
  ],
  "standardization_mean": [
    -197.50550079345703,
    -9.528732657432556
  ],
  "standardization_std": [
    71.60835921260684,
    0.4013215544248379
  ],
  "coefficients": [
    0.3958333333333333,
    0.07919995288692382,
    0.09476501404170509
  ]
},
"DT + S5": {
  "test_episode_macro_mae": 0.32063680103357006,
  "mae_95ci": [
    0.29179968402125944,
    0.35416429997275983
  ],

```

```

"test_episode_macro_spearman": 0.17435459412067472,
"valid_correlation_episodes": 8,
"raw_score_spearman": null,
"features": [
  "DT",
  "S5"
],
"standardization_mean": [
  -197.50550079345703,
  -9.386288225650787
],
"standardization_std": [
  71.60835921260684,
  0.39092901221649673
],
"coefficients": [
  0.3958333333333333,
  0.06802864237985304,
  0.12055665482775839
]
},
"paired_comparisons": {
  "S1": {
    "combo_minus_DT_mae": 0.0014163441567961732,
    "paired_episode_bootstrap_95ci": [
      -0.005771100021116012,
      0.008584022334640323
    ]
  },
  "S3": {
    "combo_minus_DT_mae": -0.00016939132034182042,
    "paired_episode_bootstrap_95ci": [
      -0.01221041481868537,
      0.01210488015498985
    ]
  },
  "S5": {
    "combo_minus_DT_mae": -0.006703732670235857,
    "paired_episode_bootstrap_95ci": [
      -0.023619395250426306,
      0.012014485736433614
    ]
  }
},
"calibration": "z-score on development frames; ridge alpha=1, unpenalized intercept; clip predictions to [0,1]; no hyperparameter selection",
"terminal_outcome_auroc": null,
"terminal_outcome_reason": "12/12 evaluation episodes successful; no failed terminal examples",
"dt_diagnostics": {
  "dt_expected_bins": {
    "min": 68.65629577636719,
    "max": 376.024658203125,
    "mean": 203.22520340813531
  },
  "dt_entropy": {
    "min": 4.74024772644043,
    "max": 6.515954971313477,
    "mean": 5.752596414751476
  },
  "dt_valid_mass": {
    "min": 0.5985760688781738,
    "max": 0.9226280450820923,
    "mean": 0.7324754769603411
  }
},

```

```

    "dt last bin mass": {
      "min": 2.1003496897264995e-07,
      "max": 3.9913338696351275e-06,
      "mean": 2.1251089726466907e-06
    }
  },
  "limitations": [
    "One task; eight test episodes; all successful demonstrations.",
    "Independent reference has different mug colors and randomized layout; scene matching is incomplete.",
    "128 fixed shuffled patches/image; controlled history comparison differs from the training uniform total-budget allocation.",
    "Ridge fits can reverse raw-score signs; raw correlation and coefficients are reported.",
    "Training membership not fully audited; no zero-shot generalization claim.",
    "DINOv2, Robometer, bottles and plates have not been run."
  ]
}

Label validation
{
  "upstream_commit": "6c467cebcecf16a4dce79e6fd87a7ca2281c3ef0",
  "source_sha256": {
    "base": "c0171ad6aaf4c46e3eb62df0d8f9b6dbaca3cca3a0b0ce9596af71cc0d964e70",
    "mugs": "f5ffaceff57d911ab698bd787705f278913fe0c9af44e85db93ad356527057cc"
  },
  "episodes": 13,
  "frames_checked": 10242,
  "result": "exact reward parity on all frames",
  "scope": "mug qpos addresses derived from assembled XML; evaluator methods executed unchanged; no physics replay"
}

Serialization validation
{
  "result": "passed",
  "input_tokens_by_history": {
    "1": 1279,
    "3": 2559,
    "5": 3839
  },
  "target_code_tokens": 512,
  "patches_per_image": 128,
  "checks": [
    "same target and current patches across arms",
    "all four target codes, pointers excluded",
    "right-shifted target index k-1",
    "goal role only for DT; goal time zero",
    "maximum input length below 4096"
  ]
}

```